Level	Phase	Cognitive control state	Grounding	occ.
L1 goal-setting	PLAN	goal decomposition; PFC sets the goal hierarchy	[2, 4]	0
L2 execution	RETRIEVE	directed, externally-directed acquisition	[5]	4,298
L2 execution	INSPECT	metacognitive monitoring; no proposition tested	[6]	4,079
L2 execution	EXTRACT	central executive locks on a target value	[7]	1,681
L2 execution	VERIFY	metacognitive control; output is a truth value	[6, 8]	7,931
L2 execution	WRITE	generative production; externalisation mode	[9]	5,824
L3 integration	SYNTHESIZE	multi-source integration into a new representation	[10]	0
L4 interrupt	REPAIR	error monitoring; SAS override after a failure signal	[11]	116
L4 interrupt	HANDOFF	distributed cognitive load transfer (agent change)	[10, 12]	0

Table 1: The nine cognitive phases over four abstraction levels, with their control-state definitions and grounding. The *occ.* column gives the per-phase occurrence count over the 500-trajectory corpus. The empty L3/L4 phases are a finding, not a gap: single-agent coding traces do not integrate across sources (SYNTHESIZE) or transfer control (HANDOFF), and overt planning (PLAN) is latent in reasoning the trace does not externalise.

1 Procedural annotation framework

We annotate each agent trajectory in three layers, following a standoff multi-layer design conformant to the Linguistic Annotation Framework (ISO 24612:2012; [1]).

Layer B·1a: action tokens. Each raw event is normalised, with *no* interpretive rule, into an *action token* carrying four fields, $a = (\textit{action\_type}, \textit{tool\_family}, \textit{agent\_id}, \textit{artifact\_id})$, and adjacent identical tokens are merged (de-duplication). This is pure data cleaning and is the substrate every downstream layer references.

Layer A: cognitive phases. Each span is assigned, *first*, one of nine cognitive control states grounded in the prefrontal-cortex theory of cognitive control [2]. The phases are organised over four abstraction levels — L1 goal-setting (PLAN); L2 execution (RETRIEVE, INSPECT, EXTRACT, VERIFY, WRITE); L3 integration (SYNTHESIZE); L4 interrupt (REPAIR, HANDOFF). The phase label is an *optional, coarse procedural descriptor* used for interpretability and diagnostic analysis; it is *not* a fixed skill type. HPOP does not take phases as skill labels: it receives the CPA occurrence sequences and learns a reusable library of *local partial-order skills*, and a learned skill may contain CPAs from one phase or several. The phase taxonomy is given in Table 1.

Layer B·1b: canonical procedural actions. Within each phase, the span is refined to a *canonical procedural action* (CPA) — the procedural *function* realising that control state, more abstract than a tool call and portable across repositories. CPAs correspond to the *Action* tier of Activity Theory [3]: goal-directed, consciously executed units, as opposed to the automatised

Operations (raw clicks/commands) at which reusable structure does not emerge. The annotation is thus coarse-to-fine: phase first, CPA within it, each CPA realising exactly one phase.

2 Evaluating the annotation against no-instruction baselines

A labelling scheme assigns a symbol $\ell \in V$ to each occurrence. A *good* scheme makes the trajectory predictable: the current label should be informative about the next. We therefore measure the sequential structure a scheme induces, relative to a null that destroys order.

2.1 Transition information and the shuffle null

For a scheme, collect the multiset of consecutive label pairs over all trajectories, $\mathcal{P} = \{(\ell_t, \ell_{t+1})\}$, and estimate the *transition mutual information*

$$\hat{I} \;=\; \sum_{x,y \in V} \hat{p}(x,y)\,\log_2 \frac{\hat{p}(x,y)}{\hat{p}(x)\,\hat{p}(y)}, \tag{1}$$

where $\hat{p}(x,y)$ is the empirical joint over adjacent pairs and $\hat{p}(x), \hat{p}(y)$ the current/next marginals. If the next label is independent of the current one, $\hat{I} = 0$.

Finite samples inflate $\hat{I}$ by an $O(|V|^2/N)$ bias, so a larger vocabulary scores spuriously higher. We remove it with a within-trajectory *shuffle null*: permute each trajectory's labels uniformly at random (preserving length and unigram composition, destroying order), recompute (1), and average over $B = 150$ permutations to obtain $\bar{I}_0$. Because the null shares the exact marginals and N of the data, $\bar{I}_0$ estimates the finite-sample floor for that scheme. We report the *excess* structure and a size-normalised *efficiency*,

$$\Delta \;=\; \hat{I} - \bar{I}_0, \qquad \eta \;=\; \frac{\Delta}{\log_2 |V|}, \tag{2}$$

where $\log_2 |V|$ is the maximum information a single symbol can carry, making schemes of different vocabulary sizes comparable. By construction a labelling with no ordering structure has $\Delta \approx 0$.

2.2 Baselines

All schemes label the *same* occurrence segmentation; only the symbol differs.

cmdword (no instruction). ℓ is the raw leading command token (`pytest`, `grep`, `view`, ...); no taxonomy. A natural, uncontrolled vocabulary.

actobj (no instruction). $\ell = (\textit{action\_type}, \textit{artifact\_role})$, e.g. `run:test`, `edit:source`; still no procedural taxonomy.

v1 cpa (induced). The bottom-up dictionary: surface forms mined from data, then grouped into phases post hoc.

v2 cpa (cognitive). The top-down dictionary of Section 1: nine phases first, CPAs derived as their realisations.

2.3 Results

Table 2 reports the ladder on 500 SWE-rebench/OpenHands trajectories (334 repositories; 23,929 occurrences). A *no-instruction* labelling is the honest floor; both principled dictionaries lie well above it, and the cognitive v2 scheme attains the highest excess and the highest per-symbol efficiency.

scheme	$\lvert V \rvert$	$\hat{I}$	$\bar{I}_0$	Δ (bits)	η
CMDWORD (no instruction)	62	0.623	0.041	0.582	0.098
ACTOBJ (no instruction)	24	0.941	0.023	0.918	0.200
v1 CPA (induced)	28	0.968	0.027	0.940	0.196
v2 cpa (cognitive)	29	1.009	0.030	**0.979**	**0.201**

Table 2: Sequential structure beyond a shuffled null (Eq. 2). The naive no-instruction labelling sprawls to $|V| = 62$ yet is half as efficient ($\eta = 0.098$); the cognitive dictionary packs the most non-random structure into the fewest reusable symbols.

Information beyond the surface. To confirm the abstraction is not a relabelling of the physical layer, we compare three *layers* of the same trace by (i) normalised predictability NP $= \hat{I}(\ell_t; \ell_{t+1})/\hat{H}(\ell_t)$ and (ii) the information a current label gives about the next physical action, IG $= \hat{H}(a_{t+1}) - \hat{H}(a_{t+1} \mid \ell_t)$. At nearly equal vocabulary the cognitive phase layer ($|V|$=6, NP=0.186, IG=0.253 b) is about twice as self-predictable as the raw physical layer ($|V|$=5, NP=0.097, IG=0.159 b), and a current phase is *more* predictive of the next physical action than the current physical action itself — the abstraction explains the surface, not vice versa.

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